

K.S.R.M. COLLEGE OF ENGINEERING (AUTONOMOUS), KADAPA
B.Tech IV Semester (R23UG) Reg. & Supple. Exams Jun. – Jul. 2025
OPERATING SYSTEMS
(CSE)

Time: 3 Hours**Max. Marks: 70****Instructions:**

1. Answer all 10 questions from Part-A. Each question carries two marks
2. Answer one full question from each unit in Part-B. Each full question carries 10 marks

PART-A				10*2 = 20M		
1	a	Differentiate between user mode and kernel mode in an OS.	2M	CO1	L3	
	b	What are the key services provided by an operating system?	2M	CO1	L2	
	c	Define a process and its different states	2M	CO2	L2	
	d	What is the difference between a process and a thread?	2M	CO2	L2	
	e	What are the four conditions necessary for deadlock?	2M	CO3	L2	
	f	What is the critical section problem?	2M	CO3	L2	
	g	What is demand paging	2M	CO4	L2	
	h	List different page replacement algorithms	2M	CO4	L2	
	i	List the different file operations	2M	CO5	L2	
	j	Differentiate between protection and security	2M	CO5	L3	
PART-B				5*10=50M		
UNIT-I						
2		Define operating system. Explain the various functions of operating systems.	10M	CO1	L2	
OR						
3		Explain the importance of system calls and discuss their types with examples	10M	CO1	L2	
UNIT-II						
4	a	Differentiate between preemptive and non-preemptive scheduling.	5M	CO2	L4	
	b	Given a set of processes, apply the Round Robin scheduling algorithm and compute average waiting time.	5M	CO2	L3	
(OR)						
5	a	Explain the role of a process control block (PCB) in process management.	5M	CO2	L2	
	b	Discuss the advantages and disadvantages of multithreading in modern systems.	5M	CO2	L2	
UNIT-III						
6	a	Explain the producer-consumer problem and solve it using semaphores.	5M	CO3	L3	

	b	Explain the deadlock avoidance with the help of Banker's algorithm.	5M	CO3	L4
(OR)					
7	a	Discuss different synchronization mechanisms like Mutex Locks and Monitors.	5M	CO3	L2
	b	Solve a given deadlock detection problem using a resource allocation graph.	5M	CO3	L3
UNIT-IV					
8		Consider the following page-reference string:0,1,2,3,0,1,2,3,0,1,2,3,4,5,6,7. How many page faults occur for the following page replacement algorithms, assuming Frame count=3 i)FIFO ii) LRU iii) OPTIMAL	10M	CO4	L3
(OR)					
9		Compare paging and segmentation memory management techniques.	10M	CO4	L4
UNIT-V					
10	a	Explain about access matrix.	5M	CO5	L2
	b	Explain different file allocation methods with examples.	5M	CO5	L2
(OR)					
11	a	What is protection? Explain goals, principles and domain of protection	5M	CO5	L2
	b	Compare different directory structures in a file system.	5M	CO5	L4
